Project Executable Files

Date	05 October 2026
Team ID	UNM-293 2026-2430
Project Name	FitBuddy - AI Fitness Plan Generator using Gemini Models
Maximum Marks	3 Marks
Member / Assigned Work	Kaviya sree.T – Project Executable Files, setup verification and run instructions

Project Executable Files

This section contains the files, dependencies, folder structure, deployment details and run instructions required for an evaluator to execute the FitBuddy application. FitBuddy is a FastAPI-based web application using Jinja2 templates, Google Gemini models and SQLite/SQLAlchemy persistence.

Work assigned to Kaviya sree.T: Prepare the executable-file submission, verify required dependencies and environment configuration, document the project structure, provide local execution steps, and record known issues or limitations.

Step 1: Submission Checklist

S.No	Item to Submit	Submitted (Yes / No / NA)
1	Complete source code (all files and folders)	Yes
2	README / Setup Guide (instructions to run the project)	Yes
3	requirements.txt / dependency file	Yes
4	Database schema / seed files (if applicable)	Yes / As applicable
5	Environment configuration file (.env.example or similar)	Yes – Gemini API key configuration
6	Deployed application URL (if hosted)	NA – Local deployment
7	APK / Executable binary (if applicable)	NA – Web application
8	Dockerfile / Containerization config (if applicable)	NA
9	Test files and test results	Yes
10	Demo video or walkthrough (if required)	To be added if required

Step 2: File / Folder Structure

FitBuddy follows a modular FastAPI architecture with application code, templates, static assets, dependency configuration and database support.

/fitbuddy-project
■■■ app/ – FastAPI application source code
■ ■■■ main.py – application entry point
■ ■■■ routes.py – route handling and application logic
■ ■■■ gemini_generator.py – 7-day workout generation using Gemini Pro
■ ■■■ gemini_flash_generator.py – nutrition/recovery tips using Gemini Flash
■ ■■■ updated_plan.py – feedback-based workout plan updates
■ ■■■ database.py – SQLite / SQLAlchemy persistence
■■■ templates/ – Jinja2 HTML templates
■ ■■■ index.html – user input form
■ ■■■ result.html – generated plan, tips and feedback
■ ■■■ all_users.html – admin/user view
■■■ static/ – CSS, images and other assets
■■■ requirements.txt – Python dependencies
■■■ .env.example – Gemini API configuration template
■■■ README.md – setup and execution instructions
■■■ SQLite database – stored user and workout-plan data

Step 3: Deployment / Access Details

Field	Details
Hosted / Deployed URL	Local: http://127.0.0.1:8000
Login Credentials (Demo)	No separate login required for the documented user workflow.
Platform / Hosting Provider	Local system – FastAPI + Uvicorn
Repository Link	GitHub repository link to be added by the team
Demo Video Link	To be added if required

Step 4: Run Instructions

Pre-requisites:
• Python and pip installed
• Internet connection for Gemini API requests
• A valid Gemini API key

Step 1: Download or clone the FitBuddy project.
Step 2: Create a virtual environment: **python -m venv fitbuddy-env**
Step 3: Activate it on Windows: **fitbuddy-env\Scripts\activate**
Step 4: Install dependencies: **pip install fastapi uvicorn jinja2 sqlalchemy python-multipart google-generativeai**
Step 5: Configure the Gemini API key in the environment configuration.
Step 6: Start the application: **uvicorn app.main:app --reload**
Step 7: Open **http://127.0.0.1:8000** in a browser.
Step 8: API testing/documentation is available at **http://127.0.0.1:8000/docs**.

Step 5: Known Issues / Limitations

S.No	Known Issue / Limitation	Workaround / Status
1	Gemini API requires a valid API key and internet connection.	Configure the API key before starting the application.
2	AI response time depends partly on the external Gemini service.	Retry the request if the service is temporarily unavailable.
3	The documented version is intended for local FastAPI/Uvicorn execution.	Run locally or deploy later to a suitable hosting provider.

Project basis: The FitBuddy specification identifies FastAPI + Jinja2 for the web application, Gemini 1.5 Pro for workout generation and plan updating, Gemini Flash for nutrition tips, and SQLite with SQLAlchemy for persistence.